

Frame It as Cases: Work That Speaks for Itself

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Track Deliverable: Week 2 Case Studies & Portfolio Framing

Lane: ML / Anomaly Detection

Deployed Project: masoomsakina.github.io/flyrank-internship-ml/

STANDING VOICE CARD (PORTFOLIO INSTRUCTION)

"Direct, plainspoken, analytical, grounded, no buzzwords."

Featured Case: Organic Traffic Decay Anomaly Classifier & Action Playbook

BEAT 1: THE PROBLEM

Organic search traffic decay rarely happens overnight; it erodes gradually across high-value content. At scale, SEO and editorial teams cannot manually audit tens of thousands of URLs every week to catch declining pages before compounding organic loss occurs. Incumbent heuristics rely on rigid rules (e.g., fixed percentage drops) that fail to separate genuine decay anomalies from standard seasonal variance—wasting valuable editorial budget on stable pages while missing high-exposure traffic declines. The goal was to engineer an auditable machine learning pipeline to prioritize high-risk decay anomalies for targeted content refresh workflows.

BEAT 2: WHAT I DID (AND DECIDED)

- **Data Contract & Leakage Prevention:** Ingested 30,000 anonymized records from the FlyRank warehouse release. To preserve privacy and prevent target leakage, I strictly excluded raw URLs, client identifiers, forward-looking traffic windows, and proxy trend features, isolating the historical baseline feature `impressions_90d`.
- **Task Framing & Transparent Baseline:** Formulated the problem as a binary classification task evaluated on F1-Score to balance false alarms against missed decay events. Established an incumbent rule baseline (`impressions ≥ 1,000` with negative trend) that scored **0.6672 F1**.
- **Model Training & Out-of-Domain Audit:** Trained a regularized Random Forest classifier. Moving beyond a naive stratified split (**0.7219 F1**), I audited the model under a **Client-Grouped Split (GroupKFold)** across 32 distinct clients, confirming honest generalization (**0.7140 F1**) free from entity-level data bleed.
- **Operational Action Playbook:** Mapped predictions to an actionable 4-tier decision queue (`RECOMMEND_REFRESH`, `MONITOR`, `NO_ACTION`), isolating 8,031 high-priority candidates and establishing a strict "No-Go" protocol prohibiting uncontrolled automated rewrites or redirects.

BEAT 3: WHAT CAME OF IT

The Random Forest classifier reliably outperformed the rule-based heuristic across all validation designs, reducing misclassification noise near the 1,000-impression threshold. The pipeline exports an auditable decision-support queue (`action_playbook_ranked.csv`) and serialized JSON receipts (`metrics.json`), equipping editorial leads with clear, data-backed prioritization rather than opaque model scores.

Evaluation Strategy / Pipeline Stage	Primary Metric (F1)	Operational Utility / Validation Integrity
Week-4 Incumbent Baseline Rule	0.6672	Rigid heuristic (impressions $\geq 1,000$ + downward trend)
Week-5 Random Forest (Stratified Split)	0.7219	Handles non-linear distributions; anchors exposure threshold
Week-6 Random Forest (Client-Grouped Split)	0.7140	Verifies out-of-domain generalization across 32 clients
Action Playbook Ranked Queue	8,031 flagged	High-priority pages routed to editorial refresh queue

Before & After Copy Diff (Generic AI vs. Voice Card)

GENERIC AI COPY (BEFORE)	EDITED IN VOICE (AFTER)
<i>"Leveraged cutting-edge state-of-the-art ensemble machine learning algorithms to synergistically revolutionize content marketing strategies and guarantee massive SEO ranking growth across all client portfolios."</i>	<i>"Trained a regularized Random Forest classifier on historical impression signals, outperforming a rule-based baseline (0.7219 vs. 0.6672 F1) to provide editorial teams with an auditable queue of decaying pages."</i>

Portfolio Bio & Contact CTA

Bio: Data Science student specializing in machine learning pipelines, anomaly detection, and leak-free feature engineering. I build transparent models and decision-support systems that prioritize honest validation over inflated benchmark metrics. Contact / CTA: Looking to audit data contracts, evaluate classification pipelines, or deploy decision-support workflows? Inspect the codebase at github.com/MasoomSakina/flyrank-internship-ml or reach out to collaborate.
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